

# Technical Document #320: Retrieval Augmented Generation - Comprehensive Overview

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

### Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Semantic search finds documents based on meaning rather than exact keywords.
- Chunking splits documents into smaller segments for embedding.